Data Preparation and Feature Engineering

These are optional lookup references, not extra lessons or homework. Open only the named section you need, try it in your notebook, and return to the issued LMS task. You do not need to read the full manuals.

Start with class materials: use the mentor-issued Telco CSV, dictionary, starter and fixed split card. Keep one project and the same coding agent. The dataset and starter files are not bundled into this PDF. Use approved teaching samples, not private customer records, in prompts or public repositories.

C1 Data Quality and Cleaning

pandas DataFrame.info and Working with missing data

Use the info examples and the missing-value detection section to inspect types and missing counts before editing. isna() does not tell you why a value is absent; a cleaning decision still needs an explanation.

pandas to_numeric and Working with text data

Look up errors="coerce" and string methods such as strip. Count the parsing failures introduced by conversion. Preserve one agreed category spelling; do not lowercase values and then compare them with the old capitalized labels.

C2 Outliers, Scaling and Normalization

StandardScaler and MinMaxScaler

Read the opening examples and fit/transform behavior. Compare raw scale, standard scale and min-max scale using the same validation rows. Scaling does not make unusual observations wrong. New min-max values can fall outside the training range.

scikit-learn common pitfalls

Read "Inconsistent preprocessing" and "How to avoid data leakage". Fit on training data, then reuse the learned transformation on validation data. Do not learn fill values or scaling from the whole file before a comparison.

C3 Categorical Encoding and Feature Construction

Preprocessing data

Open "Encoding categorical features". Distinguish a real order, such as small/medium/large, from arbitrary integer codes for unrelated categories. Use the class example rather than studying every preprocessing method.

OneHotEncoder

Look up handle_unknown and the examples. With handle_unknown="ignore", an unseen category produces an all-zero block instead of new columns. Inspect both the training categories and the transformed new row.

For constructed features: use the actual Telco dictionary and the formulas in the task, not invented source columns. Check two rows by hand and a zero denominator. Service counts need the actual service values; a high metric does not prove that a feature is available at prediction time.

C4 Feature Selection

pandas correlation matrix

Look up corr on the intended numeric columns. The assignment uses abs(corr)>0.85 for the feature-pair filter. Use training rows for the decision; correlation does not establish a cause.

mutual_info_classif

Use the class support block and inspect discrete_features and random_state. Tell the method which encoded columns are discrete; dense/sparse storage alone does not describe their meaning.

RFE

Look up n_features_to_select, support_ and ranking_. Inspect which columns remain and compare the required model outputs. You do not need to memorize the estimator API.

Working with the same agent

Follow the issued run card for one native project-rule mechanism. Ask the agent to identify the permitted files and plan before changing them. Verify that it applied the rules; the presence of a file alone is not proof. In a new chat, provide a short state note: current files, decisions, last verified result and next bounded task.

Keep the target, split and protected files unchanged during a feature comparison. Review the changed files, run the notebook yourself if the agent cannot run Colab, and save the checked version.

C5 Imbalanced Data, Data Splits and Leakage

imbalanced-learn common pitfalls

Read "Data leakage". Resample training observations only, leaving validation/test unchanged. If a sampler is part of a pipeline, use the prepared imbalanced-learn route so it is fitted within training folds.

Over-sampling

Read "Naive random over-sampling" and the opening SMOTE explanation. Compare duplication with synthetic interpolation. Plain SMOTE on encoded categories is a simplified teaching comparison, not a general recommendation for mixed customer data.

LogisticRegression class_weight and Tuning the decision threshold

Look up class_weight="balanced" and the threshold guide introduction. Use the starter to compare the assigned 0.3/0.4/0.5 thresholds on validation. Keep the final test set outside the choice; a larger accuracy alone does not answer which churn errors matter.

C6 Lab: Reproducible Preprocessing Pipelines

Pipelines and composite estimators

Open "Pipeline: chaining estimators" and "ColumnTransformer for heterogeneous data". Use the same fitted preparation for training and new inputs. Check a valid new record, an unseen category and a complete run after restarting Colab.

Cross-validation: evaluating estimator performance

Read "Computing cross-validated metrics", "K-fold" and "Stratified K-fold". Compare explicit five-fold splitters as assigned, passing the full unfitted pipeline. Fit preparation and selection separately inside each training fold; report actual mean/std, not a confidence interval.

Lab check: explain the order before asking the agent to assemble existing blocks. Keep CV inside development data and run the declared leakage example on an isolated copy. Keep the final test unopened throughout M3 for the next module. A partner may verify your run, not edit your assessed work.

Official references checked 25 October 2026. External references are optional support; required comparisons and submission outputs remain those of the issued LMS task. No new deadline is introduced.